Notebook 1 – Chapter 1.2

Axioms, Conditioning & Bayes' Theorem

10 hard, Stage-1-calibrated problems spanning Kolmogorov axioms, Boole/Bonferroni bounds, independence, Law of Total Probability, Bayes' Theorem, and the classical Bayesian traps

A. Kolmogorov Axioms & Independence

Q1 – Pairwise vs. Mutual Independence (the $n \geq 3$ trap)

Flip two fair coins. Let $A = \{\text{coin 1 is heads}\}$, $B = \{\text{coin 2 is heads}\}$, $C = \{\text{the two coins show the same face}\}$. Show A, B, C are pairwise independent but not mutually independent.

Solution. Sample space has 4 equally likely outcomes: HH, HT, TH, TT. $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{2}$.

$$\mathbb{P}(A \cap B) = \mathbb{P}(\{HH\}) = \tfrac{1}{4} = \mathbb{P}(A)\mathbb{P}(B). \checkmark$$

$$\mathbb{P}(A \cap C) = \mathbb{P}(\{HH\}) = \tfrac{1}{4} = \mathbb{P}(A)\mathbb{P}(C). \checkmark$$

$$\mathbb{P}(B \cap C) = \mathbb{P}(\{HH\}) = \tfrac{1}{4} = \mathbb{P}(B)\mathbb{P}(C). \checkmark$$

All three pairs are independent. But

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(\{HH\}) = \tfrac{1}{4} \neq \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C) = \tfrac{1}{8}.$$

So mutual independence fails: knowing any two of A, B, C actually determines the third exactly, which pairwise checks alone can never detect.

Q2 – Boole's Inequality, Tight Estimation

A risk desk tracks 6 independent failure modes, each with probability 0.1. Using only the union bound, give an upper bound on $\mathbb{P}$(at least one failure), then compute the exact value and the bound's relative error.

Solution. Boole: $\mathbb{P}\left(\bigcup_{i=1}^{6} A_i\right) \leq \sum_{i=1}^{6} \mathbb{P}(A_i) = 6(0.1) = 0.6$.

Exact (independence): $\mathbb{P}(\text{at least one}) = 1-(0.9)^6 = 1-0.531441 = 0.468559$.

Relative error of the bound: $\dfrac{0.6 - 0.468559}{0.468559} \approx 28.0\%$ – Boole's bound is always valid but can overshoot badly once individual probabilities aren't vanishingly small; it's exact only in the disjoint-event limit.

Q3 – Bonferroni's Second-Order (Tighter) Bound

For the same 6 failure modes (independent, $p = 0.1$ each), apply the Bonferroni inequality using pairwise intersections to get a tighter two-sided bound than Q2, and compare to the exact value.

Solution. Bonferroni (inclusion–exclusion truncated after two terms) gives, for any events,

$$\sum_i \mathbb{P}(A_i) - \sum_{i<j} \mathbb{P}(A_i \cap A_j) \;\leq\; \mathbb{P}\left(\bigcup A_i\right) \;\leq\; \sum_i \mathbb{P}(A_i).$$

Here $\sum_i \mathbb{P}(A_i) = 0.6$. Pairwise terms: $\binom{6}{2} = 15$ pairs, each $\mathbb{P}(A_i \cap A_j) = (0.1)^2 = 0.01$ by independence, so $\sum_{i<j} \mathbb{P}(A_i \cap A_j) = 0.15$. Lower bound: $0.6 - 0.15 = 0.45$. Combined with Q2's upper bound 0.6:

$$0.45 \leq \mathbb{P}(\text{at least one}) \leq 0.6,$$

and the exact value 0.468559 sits inside this much narrower window – each extra inclusion-exclusion term tightens the sandwich further (this is the general Bonferroni ladder).

B. Conditioning & Law of Total Probability

Q4 – Law of Total Probability, Multi-Stage Urn

Urn A has 3 red, 7 blue balls; Urn B has 6 red, 4 blue. A fair coin picks the urn; then *two* balls are drawn without replacement from that urn. Find $\mathbb{P}$(both balls red).

Solution. Condition on the coin (the natural partition):

$$\mathbb{P}(\text{both red}) = \tfrac{1}{2}\,\mathbb{P}(\text{both red} \mid A) + \tfrac{1}{2}\,\mathbb{P}(\text{both red} \mid B).$$

$$\mathbb{P}(\text{both red} \mid A) = \frac{3}{10} \cdot \frac{2}{9} = \frac{6}{90} = \frac{1}{15}.$$

$$\mathbb{P}(\text{both red} \mid B) = \frac{6}{10} \cdot \frac{5}{9} = \frac{30}{90} = \frac{1}{3}.$$

$$\mathbb{P}(\text{both red}) = \tfrac{1}{2}\left(\tfrac{1}{15}\right) + \tfrac{1}{2}\left(\tfrac{1}{3}\right) = \tfrac{1}{30} + \tfrac{5}{30} = \tfrac{6}{30} = \tfrac{1}{5}.$$

Q5 – Bayes' Theorem, Standard Form (Reversing the Urn)

In the setup of Q4, given that both drawn balls were red, find $\mathbb{P}$(Urn A was used).

Solution. By Bayes' Theorem and the pieces computed in Q4:

$$\mathbb{P}(A \mid \text{both red}) = \frac{\mathbb{P}(\text{both red} \mid A)\,\mathbb{P}(A)}{\mathbb{P}(\text{both red})} = \frac{\frac{1}{15} \cdot \frac{1}{2}}{\frac{1}{5}} = \frac{\frac{1}{30}}{\frac{6}{30}} = \frac{1}{6}.$$

Sanity check: $\mathbb{P}(B \mid \text{both red}) = 1 - \frac{1}{6} = \frac{5}{6}$, consistent with Urn B being red-heavy, so seeing two reds should strongly favor it.

Q6 – Odds Form: Sequential Independent Tests

A contract is anomalous with prior odds 1 : 499 (i.e. base rate 1/500). Two independent anomaly tests, each with likelihood ratio (Bayes factor) 18 for a "flag" result, both flag the same contract. Find the posterior probability the contract is anomalous.

Solution. Odds form lets Bayes factors multiply directly across independent evidence:

$$\text{Posterior Odds} = \text{Prior Odds} \times (\text{BF}_1) \times (\text{BF}_2) = \frac{1}{499} \times 18 \times 18 = \frac{324}{499}.$$

Convert odds $o = a : b$ to probability via $\frac{a}{a+b}$:

$$\mathbb{P}(\text{anomalous} \mid \text{both flag}) = \frac{324}{324 + 499} = \frac{324}{823} \approx 0.3937.$$

The odds form avoids ever touching a messy joint union of flag events – each test just rescales the odds by its own Bayes factor, independent of how many tests came before.

C. Classical Bayesian Traps

Q7 – Monty Fall (the Host Doesn't Know)

Standard 3-door Monty Hall, but now the host opens one of the two unchosen doors *uniformly at random* (he does not know where the car is) and it happens to reveal a goat. Should you switch, and with what win probability?

Solution. This is the classic trap: many assume the answer is still 2/3 as in standard Monty Hall, but the host's ignorance changes the conditioning. Condition on "door 2 opened, revealing a goat" with car uniformly at door 1, 2, or 3 and host picking uniformly between the two non-chosen doors (contestant picked door 1):

$$\mathbb{P}(\text{opens 2, goat} \mid \text{car at 1}) = \tfrac{1}{2}$$

$$\mathbb{P}(\text{opens 2, goat} \mid \text{car at 2}) = 0$$

$$\mathbb{P}(\text{opens 2, goat} \mid \text{car at 3}) = \tfrac{1}{2}$$

Each car location has prior $\frac{1}{3}$. By Bayes,

$$\mathbb{P}(\text{car at 1} \mid \text{evidence}) \propto \tfrac{1}{3}\cdot\tfrac{1}{2} = \tfrac{1}{6}, \quad \mathbb{P}(\text{car at 3} \mid \text{evidence}) \propto \tfrac{1}{3}\cdot\tfrac{1}{2} = \tfrac{1}{6}.$$

These are equal, so after normalizing, $\mathbb{P}(\text{car at 1}) = \mathbb{P}(\text{car at 3}) = \frac{1}{2}$ each. **Switching wins with probability 1/2, not 2/3** – the random reveal (Monty Fall) destroys the informational asymmetry that makes standard Monty Hall's host-knows-and-avoids-the-car rule so powerful.

Q8 – N-Door Monty Hall, General Formula

Generalize: N doors, one car. Contestant picks one; host (who knows the car's location) opens $N-2$ *other* doors, all goats, leaving one door besides the contestant's pick. Find $\mathbb{P}$(win by switching) as a function of N, and verify $N = 3$ reproduces 2/3.

Solution. $\mathbb{P}$(initial pick is car) $= \frac{1}{N}$, so $\mathbb{P}$(car is elsewhere) $= \frac{N-1}{N}$. The host's knowledgeable removal of $N-2$ goat doors never destroys

information (unlike Q7) – it funnels *all* of the "car is elsewhere" probability mass onto the single remaining unopened door. Hence

$$\mathbb{P}(\text{win by switching}) = \frac{N-1}{N}.$$

At $N = 3$: $\frac{2}{3}$. ✓ As $N \to \infty$, this $\to 1$: with many doors, the host's forced goat-revealing is overwhelming evidence that the one surviving door is special.

Q9 – Bertrand's Box Paradox

Three boxes: Box 1 has 2 gold coins, Box 2 has 2 silver coins, Box 3 has 1 gold and 1 silver. Pick a box uniformly at random, then draw one coin uniformly at random from it. The coin is gold. Find $\mathbb{P}$(the *other* coin in that box is also gold).

Solution. Classic trap: the naive answer "1/2" (either Box 1 or Box 3, seemingly symmetric) is wrong because drawing gold is twice as likely under Box 1 as under Box 3. Use Bayes with the partition over boxes:

$$\mathbb{P}(\text{gold} \mid \text{Box 1}) = 1, \quad \mathbb{P}(\text{gold} \mid \text{Box 2}) = 0, \quad \mathbb{P}(\text{gold} \mid \text{Box 3}) = \tfrac{1}{2}.$$

$$\mathbb{P}(\text{gold}) = \tfrac{1}{3}(1) + \tfrac{1}{3}(0) + \tfrac{1}{3}\left(\tfrac{1}{2}\right) = \tfrac{1}{3} + \tfrac{1}{6} = \tfrac{1}{2}.$$

$$\mathbb{P}(\text{Box 1} \mid \text{gold}) = \frac{\frac{1}{3}(1)}{\frac{1}{2}} = \frac{2}{3}.$$

The other coin is gold iff Box 1 was drawn, so the answer is $\boxed{2/3}$, not 1/2. (Equivalent fast shortcut: think of 3 gold-colored and 3 silver-colored coin-slots across all boxes, each equally likely to be "the one drawn"; 2 of the 3 gold-slots sit in the all-gold box.)

Q10 – Boy-or-Girl Paradox with a Base-Rate Trap

A family has exactly two children (each independently equally likely boy/girl, $\mathbb{P} = 1/2$). (a) Given "at least one child is a boy," find $\mathbb{P}$(both are boys). (b) Given instead "the elder child is a boy," find $\mathbb{P}$(both are boys). Explain why these differ.

Solution. Sample space of (elder, younger): $\{BB, BG, GB, GG\}$, each probability $\frac{1}{4}$.

(a) "At least one boy" excludes only GG, leaving $\{BB, BG, GB\}$, each equally likely given the conditioning event:

$$\mathbb{P}(BB \mid \text{at least one boy}) = \frac{\mathbb{P}(BB)}{\mathbb{P}(BB) + \mathbb{P}(BG) + \mathbb{P}(GB)} = \frac{1/4}{3/4} = \frac{1}{3}.$$

(b) "Elder is a boy" restricts to $\{BB, BG\}$:

$$\mathbb{P}(BB \mid \text{elder is a boy}) = \frac{\mathbb{P}(BB)}{\mathbb{P}(BB) + \mathbb{P}(BG)} = \frac{1/4}{1/2} = \frac{1}{2}.$$

The gap (1/3 vs. 1/2) is entirely about *which event you conditioned on*, not a probability paradox: part (a)'s condition is satisfied by two distinct unordered-looking-but-really-ordered configurations (BG and GB) that both fail to be "both boys," diluting the fraction, while part (b) pins down one specific child's sex and leaves only one free binary variable. This is precisely the base-rate-neglect trap in Jane-Street-style phrasing puzzles: the *exact wording* of the conditioning event changes which outcomes survive, and sloppy reading (treating "at least one boy" as equivalent to "a specific boy") is the single most common way candidates lose this point under time pressure.

Drill instruction: redo Q1, Q7, and Q10 closed-book – they are the three classic "your intuition is lying to you" traps (independence illusion, host-knowledge asymmetry, conditioning-event ambiguity) that Stage 1 writers draw on most often. If any takes over 90 seconds cold, that is this chapter's weak point.